Question			Answer	Marks	AOs		Guidance
11	(i)		Component of weight down the plane $4.7g\sin 60°$ Equilibrium equation $T = 4.7g\sin 60°$ $= 39.889...$ so $T = 39.9$ to 3 sf	**B1** **E1** **[2]**	**2.1** **3.3**	**AG** Award if seen Must be clear that 39.9 N is the tension and not just component of weight	
	(ii)		Resolve perpendicular to the slope N is the normal reaction between plane and block B $N = 4g\cos 25°$ Resolve up the slope $T - F - 4g\sin 25° = 0$ On the point of sliding so $F = \mu N = \mu \times 4g\cos 25°$ $\mu = \dfrac{4.7g\sin 60° - 4g\sin 25°}{4g\cos 25°} = 0.656$ to 3sf	 **B1** **M1** **A1** **M1** **A1** **[5]**	 **1.1a** **3.3** **1.1b** **3.1b** **1.1b**	 Need not be evaluated here $[\approx 35.5]$ Allow only sign errors F need not be evaluated here $[\approx 23.3]$ Do not allow for $F \le \mu N$ unless = used subsequently. FT their values. FT (notice this answer is 0.657 if 39.9 used for T)	If only values are seen used, it must be clear that the values used are friction and normal reaction.